

TRANSFORMING AMERICAN HEALTHCARE: THE DEPENDENCY MATRIX

Section 1.4: Architecture of Interdependency | Source: Transforming American Healthcare Framework (HTR Book, 2026)

CORE PILLARS	CROSS-CUTTING IMPERATIVE	INTERDEPENDENCIES	CRITICAL PATH SEQUENCE
5 (Policy, Tech, Econ, Clinical, Ops)	Equity (Justice Test: 'Is it just?')	(9 Core + 2 Feedback + 2 Constrain	5 Stages (Policy -> Tech -> Econ -> Clinical -> Ops)

FROM ↓ / TO →	Policy	Technology	Economics	Clinical	Operations
Policy	—	ENABLES Funds & authorizes the data infrastructure build	ENABLES (Critical Path) Makes payment reform binding (Acts 167/68)	—	DRIVES (Critical Path) Statutory deadlines force execution capacity
Technology	—	—	ENABLES (Critical Path) Analytics run the VBC financial model	ENABLES Data enables population health management	—
Economics	—	REQUIRES Needs data infrastructure & risk adjustment	—	DRIVES (Critical Path) Payment incentives reshape care delivery	—
Clinical	—	—	—	—	REQUIRES Care models need execution infrastructure
Operations	INFORMS ↻ Implementation data refines statutory mandate	RUNS ↻ Workforce operates data infrastructure & systems	—	—	—

DEPENDENCY TYPES & LEGEND	
ENABLES	Precondition — creates the capability or authority necessary for the target pillar to operate.
DRIVES	Active Force — actively reshapes behavior, structures, or incentives in the target pillar.
REQUIRES	Operational Prerequisite — target pillar cannot function or sustain without the source pillar's infrastructure.
↻ Feedback Loop	Self-Correction — operational data and workforce feedback running after initial system launch.
Critical Path	Non-Negotiable Build Order — Policy (Gate 1) → Tech (Gate 2) → Econ (Gate 3) → Clinical (Gate 4) → Ops (Gate 5).
Equity Imperative	Cross-Cutting Plumb Line — Justice test ('Is it just?') applied unconditionally to every cell and relationship.

DETAILED INTERDEPENDENCY CATALOG (FIGURE 1.3 BREAKDOWN)

Complete breakdown of directed dependencies, mechanisms, failure modes, Vermont evidence, and equity tests.

ID	Source Pillar	Target Pillar	Verb / Type	Critical Path?	Structural Mechanism / Description	Failure Mode When Missing ('What Breaks')	Vermont Evidence & Application	Equity Imperative Test ('Is It Just?')
DEP-01	Policy	Economics	ENABLES	Yes	Mandatory statutory authority converts voluntary reform into binding requirements, eliminating opt-outs.	Voluntary VBC contracts fail as high-cost, profitable actors opt out, causing adverse selection.	Acts 167 & 68 mandate Reference-Based Pricing (FY27) and Global Hospital Budgets (FY28).	Does the statutory mandate close regional and safety-net disparities or penalize safety-net providers?
DEP-02	Policy	Operations	DRIVES	Yes	Statutory deadlines act as organizational forcing functions that compel project execution capacity.	Transformation planning remains perpetually 'in progress' without actual execution or accountability.	Act 68 statutory deadlines: FY27 RBP, FY28 Global Budgets, Dec 2028 Statewide Strategic Plan.	Are operational deadlines enforced equitably across rural critical access and urban health centers?
DEP-03	Policy	Technology	ENABLES	No	Legislation authorizes, mandates, and funds state-wide health data infrastructure and HIE integration.	Health information exchange remains voluntary, fragmented, and underfunded across community sites.	Act 62 transfers HIE authority to DVHA; \$195M/yr RHT award funds CIN & technology upgrades.	Does state technology funding prioritize data infrastructure in historically under-resourced rural counties?
DEP-04	Technology	Economics	ENABLES	Yes	Data analytics substrate enables total cost of care (TCOC) tracking, attribution, and risk adjustment.	'Managing Blind' failure mode: assuming financial risk without real-time analytics produces unexpected losses.	VHCURES all-payer claims database & joint AHS-GMCB analytics vendor procurement.	Does risk adjustment incorporate social risk factors (SDOH) to protect safety-net providers?
DEP-05	Technology	Clinical	ENABLES	No	Data infrastructure enables population health management, risk stratification, and care gap tracking.	Care management operates reactively after acute hospitalizations rather than proactively before crisis.	Blueprint clinical data registry & VITL/VHIE clinical exchange driving 5.8:1 primary care ROI.	Is clinical data disaggregated by race/ethnicity to expose disparities hidden in aggregate averages?
DEP-06	Technology	Equity	ENABLES	No	Demographic data stratification reveals disparities hidden in population-level aggregate averages.	Aggregate performance appears strong (e.g. 91% primary care access) while severe subgroup gaps persist.	VHCURES stratification revealing 11-point primary care access gap between white and BIPOC Vermonters.	Are demographic fields complete across commercial, Medicare, and Medicaid claims data?
DEP-07	Economics	Clinical	DRIVES	Yes	Payment incentives flip financial logic from volume (FFS) to population health management (VBC).	Clinical care redesign fails under FFS because preventing hospitalizations cuts provider revenue.	Global hospital budgets & Blueprint PMPM payments aligning financial and clinical logic.	Do payment incentives reward serving high-SDOH, hard-to-reach populations?
DEP-08	Economics	Technology	REQUIRES	No	Value-based contracts cannot be executed, monitored, or disputed without claims/EHR analytics.	CFOs sign risk contracts without analytics to verify or dispute payer shared-savings reconciliations.	Act 68 requiring GMCB to set prospective global budgets using VHCURES historical data.	Are contract benchmarks adjusted for social risk in high-poverty counties (e.g., Essex County)?
DEP-09	Clinical	Operations	REQUIRES	No	Care delivery models (PCMH, CoCM, CCBHC) require workforce, credentialing, and management capacity.	Care models remain paper diagrams without operational workforce and administrative workflows to run them.	Collaborative Care Model (CoCM) requiring BH care managers and telepsychiatry infrastructure.	Is care model expansion executed in rural and under-resourced communities?
DEP-10	Operations	Policy	INFORMS ↻	No	Implementation data feeds back into policy design, creating self-correcting policy learning loops.	Top-down statutory mandates fail to adapt to real-world operational and administrative bottlenecks.	AHS monthly transformation reports to legislature refining Act 68 implementation rules.	Does operational feedback from rural providers directly shape statutory adjustments?
DEP-11	Operations	Technology	RUNS ↻	No	Human workforce (IT staff, data analysts, privacy officers) operates and maintains data platforms.	Sophisticated technology platforms become unused 'shelfware' due to lack of staff execution capacity.	Vermont CIN shared services model providing IT/analytics staff to rural CAHs.	Is IT and analytics workforce capacity distributed equitably to resource-thin rural hospitals?
DEP-12	Equity	Policy	CONSTRAINS	No	Justice reviews constrain policy design to ensure statutory mandates do not widen health disparities.	Policies designed without equity input penalize safety-net hospitals serving high-poverty populations.	Act 68 requiring Health Equity Advisory Commission consultation on the Strategic Plan.	Is equity review embedded at the start of policy design rather than at approval?
DEP-13	Equity	Clinical	CONSTRAINS	No	Cultural competency and implicit bias reduction constrain care delivery to ensure equal care quality.	Expanding care delivery into underserved areas perpetuates implicit bias and cultural friction.	Blueprint universal HRSN screening & culturally competent care protocols.	Are care models effective for all demographic and geographic subpopulations?

THE FIVE-STAGE EXECUTION SEQUENCE & FAILURE CASCADES

Why order is architecture: Policy → Technology → Economics → Clinical → Operations

THE 5-STAGE EXECUTION SEQUENCE				
Stage #	Pillar	Key Action / Focus	Why This Sequence (Dependency Logic)	Vermont Statutory Anchor
Stage 1	Policy	Establish enforceable statutory authority & secure capital	Nothing can be authorized, built, or funded without statutory permission. Mandatory architecture prevents opt-outs.	Acts 167, 51, 68; AHEAD Agreement; \$195M/yr RHT Award
Stage 2	Technology	Build integrated clinical + claims data substrate in real-time	Technology is the operating system. Analytics must be live BEFORE financial risk is assumed (OneCare lesson).	VHCURES, VITL/VHIE, AHS-GMCB Analytics Vendor, Vermont CIN
Stage 3	Economics	Deploy global budgets & reference pricing onto visible system	Economics-as-management requires the technology substrate. Payment reform creates financial logic for value.	Global Hospital Budgets, Reference-Based Pricing (RBP <=200% Medicare)
Stage 4	Clinical	Redesign care delivery models on aligned incentives	Clinical redesign is where ROI is realized — but only when payment incentives reward prevention over volume.	Blueprint PCMH (5.8:1 ROI), CoCM Behavioral Health, CCBHC Expansion
Stage 5	Operations	Close administrative cost gap & execute operating plans	Operations translates strategy into reality. Admin cost gap (\$1,303/discharge) is primary source of direct savings.	RHRC Methodology, 14-Hospital Transformation Plans, AHS Restructuring

FAILURE CASCADE ANALYSIS: WHAT BREAKS WHEN A PILLAR IS ABSENT				
Pillar Removed	Immediate Failure Mode	Second-Order Failure	System-Level Outcome	Historical / Real-World Example
Policy Missing	Payment reform remains voluntary; highest-cost, most profitable actors opt out.	Clinical programs continue without financial alignment; technology built but unused for risk management.	Incremental reform among willing actors; no system change; hospital financial crisis deepens.	Vermont 2010–2022: OneCare voluntary ACO model (9/14 hospitals in losses by 2023).
Technology Missing	'Managing Blind': VBC contracts signed without real-time analytics or attribution.	Contracts produce unexpected losses; clinical care managers cannot identify high-risk patients before acute crisis.	VBC produces losses for providers; providers exit VBC; payment reform stalls; equity gaps invisible.	Most early CMMI ACO failures: organizations entered downside risk without analytics.
Economics Missing	Clinical quality programs run under fee-for-service incentives.	Preventing hospitalizations cuts hospital revenue; 5.8:1 Blueprint ROI accrues to payers, not providers.	Quality improves at the margin, but costs keep rising; financial case for transformation collapses.	US national experience: 20+ years of quality improvement under FFS without payment reform.
Clinical Missing	Payment reform changes financial incentives but has no care models to reduce utilization.	Operations manages global budget caps through blunt service cuts rather than care redesign.	Hospital financial crisis accelerates; rural hospitals close beds without outpatient alternatives.	Maryland early period: global budgets without adequate primary care/SUD investment.
Operations Missing	Statutory mandates exist on paper; clinical care models designed; data platforms bought.	Policy accountability fails without capacity to meet deadlines; platforms deployed but unused.	Transformation reports accumulate on shelves; statutory deadlines slip; deficit trajectory continues.	Most state health reform efforts of 2010s: correct policy, adequate financing, zero execution capacity.

THE PILLAR DIAGNOSTIC FRAMEWORK & EQUITY IMPERATIVE

Diagnostic questions, structural roles, outputs, and cross-cutting equity tests.

Pillar / Domain	Diagnostic Question	Structural Role in System	What It Produces When Functioning	Equity Imperative Test ('Is It Just?')
Policy	Is it permissible?	The Mandatory Architecture. Converts aspirational reform into binding requirements.	Statutory mandates eliminating opt-outs; non-deferrable timelines; binding state-federal agreements.	Permissible — and does the mandate close disparities or widen them for safety-net hospitals?
Technology	Is it possible?	The Data Substrate. Makes population health management and equity measurement analytically feasible.	Real-time attribution, TCOC dashboards, HEDIS stratification, risk prediction, AI workflow automation.	Possible — and does the data make disparities visible, or bury them in aggregate averages?
Economics	Is it sustainable?	The Incentive Architecture. Determines whether organizations have financial reason to behave differently.	Payment models making population health financially rational; reference pricing dismantling cross-subsidy traps.	Sustainable — and do the incentives reward serving the hardest-to-reach, or penalize it?
Clinical	Is it effective?	The Mechanism of Change. Translates payment incentives into actual patient health outcomes.	Team-based primary care preventing hospitalizations; behavioral health integration; chronic disease control.	Effective — and effective for whom? Are outcomes equal across demographic groups?
Operations	Is it executable?	The Execution Layer. Translates statutory mandates and care models into implemented reality.	Executed hospital transformation plans; regionalization transfer networks; PMO tracking statutory deadlines.	Executable — and executable everywhere, including in rural and under-resourced settings?
Equity Imperative	Is it just?	Cross-Cutting Plumb Line. The justice test held against every pillar at every stage.	Systematic elimination of access, quality, and outcome gaps across race, income, and geography.	The master test that Policy, Tech, Econ, Clinical, and Ops must all pass individually.